

Dual-task physiological learning for radar-based continuous blood pressure monitoring: classification-regularized regression

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ABSTRACT

Continuous blood pressure (BP) monitoring is critical for hypertension management, yet conventional non-contact radar systems face challenges such as feature space overlap under low signal-to-noise ratio conditions and inaccurate estimation during physiological state transitions due to the susceptibility of millimeter-level cardiovascular vibration signals to environmental interference. To address these challenges, we propose a dual-task learning model integrating classification-constrained regression with multi-scale spatiotemporal feature extraction. Our framework combines: (1) A hybrid ResNet-BiGRU backbone capturing waveform morphology and hemodynamic continuity through multi-scale convolutions (kernels = 15/7/3) and triple-layer bidirectional gating, enabling robust **2-second-interval predictions**; (2) A physiological regularization mechanism where classification-derived BP-range probabilities (10 mmHg bins) constrain regression outputs, suppressing implausible fluctuations during dynamic states.

Validated on 30 subjects across resting, Valsalva, and tilt-table tests, results indicate clinically relevant accuracy (SBP: -0.21 ± 6.74 mmHg; DBP: 0.25 ± 4.81 mmHg) at 0.5 Hz sampling rate, while demonstrating improved dynamic-state performance versus benchmarks with DBP RMSE reductions up to 8.4 % by dual-task strategy. This work suggests dual-task learning can mitigate radar-specific SNR constraints and physiological nonstationarity while fulfilling clinical real-time monitoring demands (beat-to-beat resolution), contributing to practical deployment of cuffless BP devices.

1. Introduction

Hypertension affects 1.3 billion adults globally, with half undiagnosed and contributing to over 10 million annual deaths [1,2]. This silent epidemic necessitates continuous blood pressure (BP) monitoring for early detection and effective management.

Conventional BP monitoring faces inherent limitations: Invasive procedures carry risks of trauma and infection [3], cuff-based methods can interfere blood flow and cause discomfort [4,5], cuffless contact-based sensors (e.g., PPG/bioimpedance [6–12]) suffer from compromised signal integrity and dermatological risks during prolonged wear due to perspiration-induced interface instability and epidermal compromise [13]. Skin-contact sensors impose additional burdens on patients, including dermatological compromises and movement restrictions, which hinder patient-centered long-term monitoring.

Radar technology overcomes skin-contact limitations by enabling non-invasive detection of cardiovascular pulsations via millimeter-scale skin vibrations [14–17]. Radar's robustness against environmental factors and individual attributes, such as lighting, temperature, skin tone, and perspiration, enhances its reliability and versatility, making it a promising solution for long-term BP surveillance. However, this contact-free paradigm introduces the fundamental signal processing challenges: Sub-millimeter arterial motions coupled with millimeter-level skin displacements are susceptible to contamination by motion artifacts and environmental interference, degrading signal fidelity. Low-SNR conditions cause feature space overlap between BP states, while physiological nonstationarity – feature distribution shifts during physiological transitions (rest-to-exercise, respiratory modulation) – complicate precise BP estimation model, highlighting the need for advanced algorithms.

To overcome the limitations of contact-based BP monitoring,

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researchers have developed regression-based radar estimation algorithms. However, two persistent challenges merit further attention: (1) balancing real-time resolution with noise robustness remains difficult for reliable beat-to-beat tracking, and (2) the absence of physiological boundary constraints in current frameworks may compromise prediction plausibility during dynamic physiological states.

To transcend this limitation, we propose a dual-task learning architecture exploiting physiological priors. The main contributions of this work are as follows:

- ① Multi-scale feature extraction resilient to low-SNR conditions. We propose a dual-branch multi-scale ResNet that concurrently optimizes regression and classification objectives. This architecture extracts hierarchical waveform features through parallel convolutional pathways, capturing both local pulsatile details and global morphological trends to enhance feature robustness under low-SNR conditions.
- ② Bidirectional temporal modeling capturing continuity priors. The dual-task learning architecture processes 2-second radar segments to achieve 0.5 Hz beat-to-beat BP estimation, explicitly enforcing hemodynamic continuity priors (cardiovascular inertia enforces beat-to-beat BP changed smoothness) through sequential pulse correlation. This design suppresses physiologically implausible fluctuations while providing continuous constraints for regression outputs, transforming fragmented predictions into clinically viable waveforms.
- ③ Classification probability fusion enforcing physiological bounds. The framework integrates BiGRU temporal modeling with classification-derived probabilities priors. BiGRU layers capture hemodynamic continuity constraints across sequential pulses, while fused classification probabilities provide physiological bounds to regularize regression outputs, suppressing non-physiological predictions.

The paper is structured as follows: Section 2 reviews related work on non-contact BP monitoring. Section 3 details the radar signal preprocessing methodology. Section 4 introduces the proposed dual-task learning architecture. Section 5 presents results and performance analysis. Section 6 provides comparative evaluation, ablation studies, and discusses limitations. Section 7 concludes with findings and future directions.

2. Relate work

Current radar-based BP estimation predominantly adopts regression-

based frameworks, as shown in Table 1, implemented through three approaches: traditional statistical regression, feature-engineered machine learning, and deep learning architectures.

Traditional statistical regression. Statistical regression methods leverage hemodynamic principles like the Moens-Korteweg equation to estimate BP by modeling pulse wave features (e.g., PTT/PWV), offering physiologically interpretable models with low computational complexity and simple calibration suitable for resource-limited devices [16,18,23,28]. However, these approaches are fundamentally constrained by oversimplified physiological assumptions—including static vessel viscoelasticity and geometry that ignore arterial compliance dynamics—along with vulnerability to radar-specific artifacts where signal quality issues cause fiducial point misalignments in pulse waveforms, critically compromising estimation accuracy. Furthermore, they fail to decouple the confounding interplay between BP and heart rate modulation during physiological activities, rendering them limited in applicability to dynamic scenarios.

Machine learning. Machine learning methods advance radar-based BP monitoring by establishing data-driven nonlinear mappings between engineered features—such as time-domain indices (e.g., diastolic duration ratio), spectral components, and pulse propagation parameters—and BP [19,21,22,24–27]. Techniques like breath-holding protocols artificially stabilize signals to enhance feature consistency [19], while multi-site fusion enriches feature spaces [21]. Despite improved adaptability through feature engineering, these methods suffer from suboptimal feature comprehensiveness—where handcrafted descriptors fail to capture complex hemodynamic interactions—and operator-dependent variability in feature selection/extraction, collectively constraining achievable accuracy.

Deep Learning Approaches. Deep learning methods employ end-to-end architectures to learn spatiotemporal representations directly from raw radar signals, circumventing limitations of handcrafted feature design [20,29,30]. Typical frameworks utilize convolutional neural networks (CNNs) [20], residual networks (ResNet) [30], and deformable convolutional networks [31] to extract hierarchical features from time-frequency representations or derived spectrograms, establishing nonlinear mappings between signals and BP values. For instance: H Zhao et al. applied ResNet to radar waveforms, achieving accuracy improvements through large-scale training [30]; Ye Qiu et al. developed Stacked Deformable CNNs for enhanced feature learning [31]; S Wang et al. combined STFT spectrograms with U-Net architectures for feature extraction [32].

Current deep learning methods depend on discretely labeled BP data (2–5 min intervals) [29,30] or extended temporal windows (15 s in [31] and 10 s in [32]), restricting models to coarse-grained estimation that

Table 1
Summarized of relate work.

Time	Method	Subjects	Radar System	Label Type	Detection Site	Experimental Scenario
2018	Linear Regression [18]	3	Digital-IF CW	Discrete	Chest	Resting, Breath-holding
2021	SVM & bagging [19]	8	CW	Discrete	Chest	Breath-holding
2022	CNN [20]	20	FMCW	Discrete	Wrist	Resting
2022	Random Tree & Forest [21]	26	CW	Continuous	Chest	Resting
2022	ANN [22]	27	CW	Discrete	Chest/Wrist	3 motion states
2022	Linear Regression [23]	7	CW	Continuous	Elbow	Resting
2022	Linear Regression [16]	7	Dual-band CW	Discrete	Wrist/Back	Resting
2022	Random Tree [24]	25	CW	—	Chest	Resting
2023	Random Tree [25]	30	CW	Continuous	Chest	Resting
2023	ANN [26]	5	FMCW	Discrete	Chest/Wrist	3 motion states
2023	ANN [27]	55	CW	Continuous	Wrist	Resting
2023	Linear Regression [28]	27	FMCW	Continuous	Chest/Wrist	Resting, Breath-holding
2023	CNN & GRU [29]	36	IR-UWB	Discrete	Chest	Resting
2024	ResNet [30]	100	LFMCW	Discrete	Neck	Resting
2024	SDCN [31]	30	CW	Continuous	Chest	5 motion states
2025	Unet [32]	26	CW	Continuous	Chest	5 motion states

Continuous wave: CW; Frequency Modulated Continuous Wave Radar: FMCW; Impulse Radio ultra-wideband: IR-UWB; Support vector machine: SVM; Artificial Neural Network: ANN; Gate Recurrent Unit: GRU; Stacked Deformable Convolution Network: SDCN.

misses essential hemodynamic transients. This sparse supervision creates two core problems:

Real-Time Limitation: Expanded windows reduce sampling rates for noise robustness, preventing beat-to-beat monitoring despite its clinical necessity.

Physiological Constraint Violation: Wide windows ignore hemodynamic continuity (BP smoothness from cardiovascular inertia), enabling physiologically implausible predictions.

Besides, existing frameworks exclusively pursue regression objectives, overlooking opportunities to leverage auxiliary physiological information embedded in radar signals that could enhance estimation accuracy. This motivates our exploration beyond pure regression: Integrating classification-derived physiological priors with regression offers a promising pathway to break through the accuracy ceiling of current radar-based BP monitoring.

3. Signal preprocessing

3.1. Dataset

We utilized synchronized physiological data from the Hamburg University of Technology's Institute of High-Frequency Technology [33], ethically approved by Friedrich-Alexander-Universität Erlangen-Nürnberg (No. 85_15B). Data from 30 subjects included radar signals (2000 Hz sampling, 24 GHz CW) and continuous BP measurements (200 Hz). The Data collected during resting states, Valsalva maneuvers, breath-holding, and tilt-table tests (upright/supine), enabling algorithm evaluation under diverse BP fluctuations.

3.2. Radar detection

A 24 GHz CW radar system [34] transmitted unmodulated waves toward the thorax, with reflected signals captured by Rx antennas. The instantaneous displacement $d(t)$ relates to phase shift $\varphi(t)$ and wavelength λ by:

$$d(t) = \frac{\lambda}{2} \frac{\varphi(t)}{2\pi} \quad (1)$$

Raw I/Q components were digitized via ADC, corrected through ellipse reconstruction, and processed with arctangent demodulation to derive precise phase differences [35].

3.3. Signal preprocessing

The radar signals, initially processed through arctangent demodulation, underwent several preprocessing steps to prepare them for use in the BP detection model. To address the challenge posed by respiratory and body motion signals, which typically exhibit higher amplitudes than pulse waves, a fourth-order bidirectional band-pass filter (0.5–20 Hz) was applied to the radar echo signals. This step was essential for isolating pulse-related components and reducing noise interference.

This step was essential for isolating pulse-related components and reducing noise interference.

To ensure data reliability, outlier removal procedures were implemented by discarding signals with systolic blood pressure (SBP) values exceeding 180 mmHg or diastolic blood pressure (DBP) values below 40 mmHg. Peak detection algorithms were employed to accurately identify SBP and DBP values in the dynamic BP readings [36].

Filtered radar signals were segmented into 2-second intervals using a sliding window technique, ensuring a balance between simplicity and signal diversity. A broader band-pass frequency range preserved cardiac micro-vibrations but risked retaining low-frequency DC components. To mitigate this, we applied trend decomposition post-segmentation:

$$x_i = \beta_0 + \beta_1 t_i + \xi_i \quad (2)$$

Detrending yielded:

$$d(x) = x - (\beta_0 + \beta_1 t) \quad (3)$$

Then, L_2 -normalization standardized features for model input, ensuring magnitude-invariant processing.

Upon completion of the preprocessing phase, the distribution of various motions within the dataset was analyzed, as shown in Fig. 1. The data were partitioned into training, validation, and test sets: 65 % of the dataset was allocated for training, 15 % for validation, and 20 % for testing. During the training process, the test set was kept separate from the training and validation sets to ensure an unbiased evaluation.

4. Dual-task learning architecture

The dual-task learning architecture for BP estimation, proposed in this paper, is illustrated in Fig. 2. The BP estimation model integrates a ResNet network, a BiGRU layer, feature combination, and a regressor. Notably, the ResNet network serves a dual role: it functions as both a component of the regression network and a classification network to obtain the categorical distribution probabilities of radar signals.

4.1. Backbone network for regression

The backbone network for the regression consists of the ResNet and BiGRU networks. Fig. 3 illustrates the architecture of the multi-scale ResNet model proposed in this study. ResNet has revolutionized deep learning by addressing the vanishing and exploding gradients, as well as the degradation problem associated with increasing network depth [37]. It has proven to be both effective and efficient in image classification tasks. In this paper, The ResNet network with multi-scale kernels is used to extract regression and classification features from one-dimensional arterial pulse wave signals.

The residual unit is the fundamental building block of ResNet and can be formally defined as:

$$y = F(x, \{W_i\}) + x \quad (4)$$

where x and y are the input and output vectors of each layer, respectively, and $F(x, \{W_i\})$ represents the residual mapping to be learned.

In the regression task, the ResNet network is employed to extract spatial features from the waveform. We utilized convolutional layers with kernel sizes of 15, 7, and 3 to capture features at different scales within the radar pulse signals. Small convolutional kernels (3×1) are used to capture subtle local variations. Larger convolutional kernels (7×1 , 15×1) are employed to capture broader waveform changes. The use of multi-scale convolutional kernels allows the network to effectively capture both local and global features, thereby enhancing the model's ability to represent the complex patterns present in the radar pulse signals.

After extracting spatial features from the waveform using the ResNet network, we employ a BiGRU network to capture the temporal

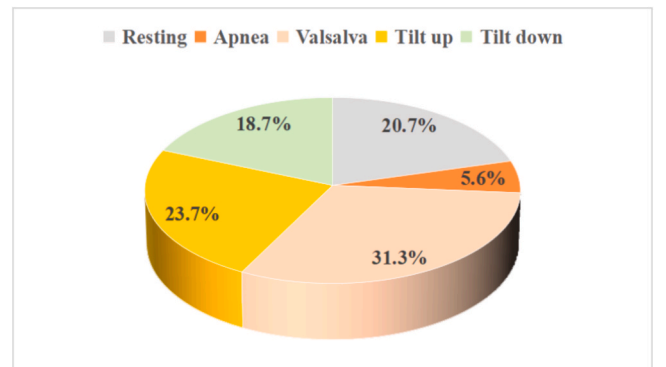


Fig. 1. The distribution of motions within the dataset.

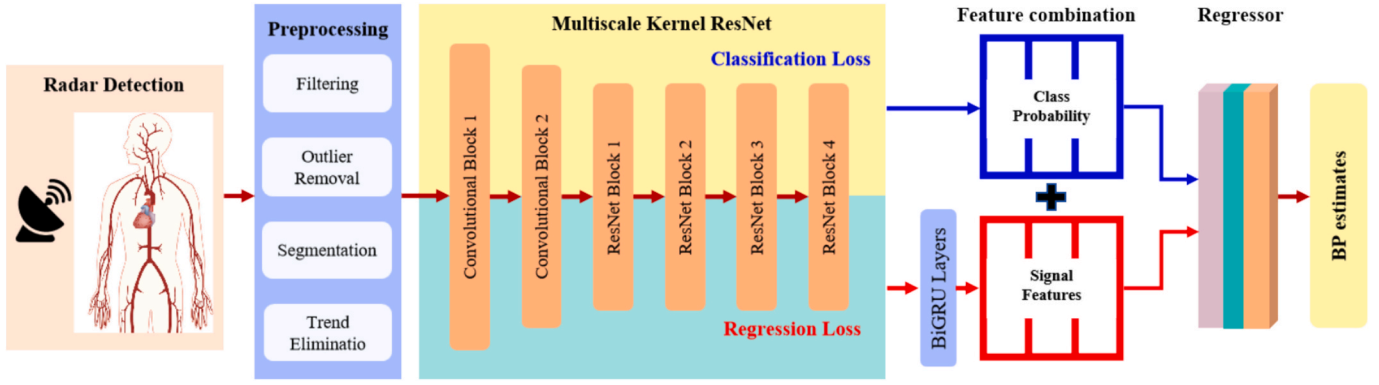


Fig. 2. Architecture of dual-task learning model.

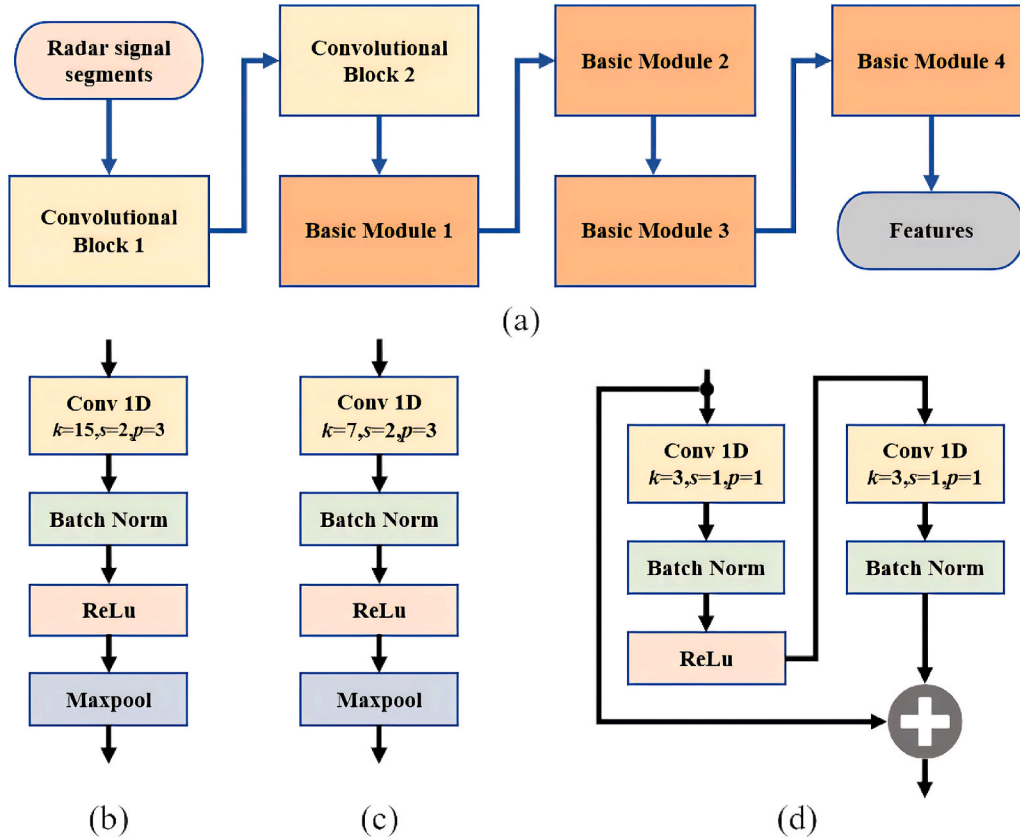


Fig. 3. (a)Architecture of multi-scale ResNet network; (b)Structure of convolution block 1; (c)Structure of convolution block 2; (d)Structure of Residual block.

dependencies in the radar pulse wave signals. The GRU is an advanced recurrent neural networks (RNNs) variant that addresses the vanishing gradient problem in long sequences [38]. Compared to traditional Long Short-Term Memory (LSTM) networks [39], GRUs achieve this with a more streamlined architecture, resulting in faster computation and training times. The BiGRU model consists of two independent GRU units: one processes data in the forward direction along the time sequence, while the other processes the data in the reverse direction [40]. The BiGRU structure is shown in Fig. 4. This bidirectional structure comprises the BiGRU model to capture both forward and backward information in the sequence, thereby enhancing its ability to understand and predict patterns within the data.

In the context of time series analysis, particularly for physiological signals, BiGRUs excel by efficiently managing long-term dependencies, making them an ideal choice for such applications. In our study, we

configured the BiGRU network with three layers. To prevent overfitting, we applied L2 regularization to the weights of the BiGRU layers, which helps to smooth the learning curves and improve generalization.

For regression, we used MSE as the loss function. MSE is a widely recognized and effective metric for evaluating the performance of regression models. It calculates the average squared difference between the predicted values and the actual values, thereby penalizing larger errors more heavily due to the squaring operation. The formula for MSE is given by:

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (5)$$

where N is the number of data points, y_i is the actual value, and $\hat{y}_i$ is the predicted value.

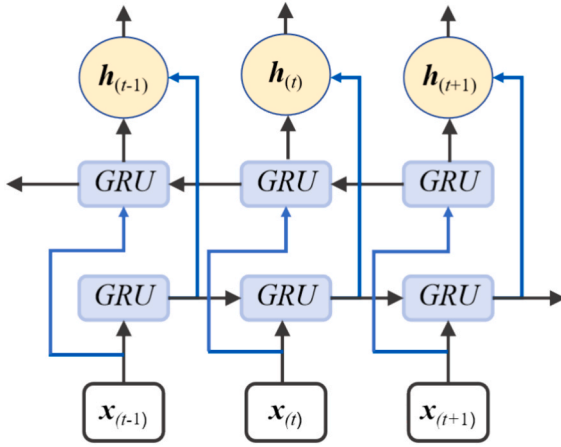


Fig. 4. The structure of BiGRU network.

4.2. Backbone network of classification

In the proposed dual-task learning model for BP prediction, a classification network is used to extract feature information from radar pulse signals from a classification perspective. This approach enhances BP prediction performance by capturing additional insights that are not easily attainable through regression alone. By dividing the BP range of 40–180 mmHg into categories at 10 mmHg intervals, we can effectively capture the distinct characteristics of different BP levels, which improves the model's ability to generalize and manage variations in the data.

The category division formula is as follows:

$$\text{Category} = \lfloor \frac{y - 40}{10} \rfloor + 1 \quad (6)$$

where y is the value of BP, $\lfloor \cdot \rfloor$ denotes the floor function, which rounds down to the nearest integer. This formula maps BP values to categories ranging from 1 to 14, corresponding to the intervals [40–50), [50–60), ..., [170–180] mmHg.

The classification task also employs the proposed multi-scale kernel ResNet network as the backbone, consistent with the regression task, as shown in Fig. 3. The multi-scale kernel ResNet network is designed to capture features at multiple scales, thereby enhancing the model's capacity to handle complex patterns and variations in radar pulse signals and improving overall performance. Sharing the ResNet network between the two tasks ensures consistency in feature extraction, leading to more coherent and reliable BP predictions.

Since BP values are continuous, using a 10 mmHg interval for class labels may result in some radar signal segments exhibiting feature distributions closer to their adjacent classes. In this study, the classification results are leveraged to enhance the accuracy of BP prediction. Therefore, in the classification task, we treat signals classified into adjacent classes as correct, as they still provide useful constraints for estimating the BP range.

To address this, we constructed a custom loss function for the classification task as follows:

$$L_{\text{class_adj}} = L_{\text{base}} + \lambda L_{\text{adj}} \quad (7)$$

where λ is the weight coefficient, L_{base} is the base cross-entropy loss function [41], defined as:

$$L_{\text{base}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(p_{i,c}) \quad (8)$$

Here, C is the number of classes, $y_{i,c}$ is an indicator function (0 or 1) that equals 1 if the true class of sample i is c and 0 otherwise, and $p_{i,c}$ is the predicted probability of sample i belonging to class c .

L_{adj} is a reward term we provide when the predicted class is an adjacent category to the true label. First, we define pred_class_i as the predicted class for sample i , true_class_i as the true class for sample i .

For adjacent classes, we provide a reward a :

$$R_{\text{adj}} = \begin{cases} a & \text{if } |\text{pred_class}_i - \text{true_class}_i| = 1 \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

The term L_{adj} can be expressed as:

$$L_{\text{adj}} = -\frac{1}{N} \sum_{i=1}^N R_{\text{adj},i} \quad (10)$$

where N is the number of samples.

4.3. Feature combination and BP estimation

The sample class probabilities obtained from the classification task are combined with the waveform and temporal dependency features extracted by the regression network to form the input for the regressor. The regressor is constructed to establish the mapping between signal features and BP readings.

By integrating the class probabilities from the classification task with the features extracted by the regression network, the model gains a multi-perspective understanding of the data. This integration enhances the feature set, providing more comprehensive information for the regression task. The additional features obtained from the classification loss function serve as a form of regularization. Helping prevent overfitting in the regression task by introducing constraints that encourage the model to generalize better to unseen data.

The classification and regression networks employ modified ResNet architectures with task-specific output heads. Detailed layer-by-layer specifications for both networks, including configuration parameters and dimensionality progression, are documented in Appendix A (Tables A1 and A2). Corresponding hyperparameter configurations are systematically presented in Tables A3 and A4.

5. Results

5.1. Performance evaluation

In this study, we evaluated the model's performance using a set of metrics: Mean Error (ME), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). These metrics offer a comprehensive assessment of predictive accuracy:

$$ME = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i) \quad (11)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (12)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (13)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\% \quad (14)$$

where n is the number of samples, y_i is the ground truth, $\hat{y}_i$ is the model prediction value.

5.2. Performance of classification network

The results of the classification network are summarized in Table 2. Due to the small interval between categories, the prediction accuracy is not very high. However, by expanding the valid prediction results to

Table 2
Results of Classification.

	Training		Validation		Test	
	Prediction accuracy	Adjacent class ratio	Prediction accuracy	Adjacent class ratio	Prediction accuracy	Adjacent class ratio
SBP	70.37	24.13	57.03	31.95	54.01	33.99
DBP	75.15	21.00	62.94	29.89	61.01	31.47

include adjacent categories and adding a reward term, the proportion of signals classified into adjacent categories increased. The overall probability of predictions being either the same as or adjacent to the true labels is 88.98 % for the SBP validation set and 88 % for the SBP test set. For the DBP validation set, this probability is 92.83 %, and for the DBP test set, it is 92.48 %. The high level of accuracy on both the validation and test sets demonstrates the model's robustness and generalization capabilities.

The confusion matrices for the classification network on the training set, validation set, and test set are shown in Fig. 5.

For SBP: The prediction range is from 70 mmHg to 170 mmHg, spanning 10 category labels. The x-axis represents the predicted categories, labeled from 0 to 9, while the y-axis represents the true category labels, also labeled from 0 to 9.

For DBP: The prediction range is from 40 mmHg to 120 mmHg, spanning 8 category labels. The x-axis represents the predicted categories, labeled from 0 to 7, while the y-axis represents the true category labels, also labeled from 0 to 7.

In these matrices, the x-axis and y-axis provide a clear visualization of the model's performance. High values along the diagonal indicate accurate predictions, while off-diagonal values show the frequency of predictions in adjacent categories.

5.3. Performance of BP estimation

In this study, radar signals were utilized with each sample segment lasting 2 s. The performance metrics for SBP and DBP estimation using the proposed model are summarized in Table 3. The model was

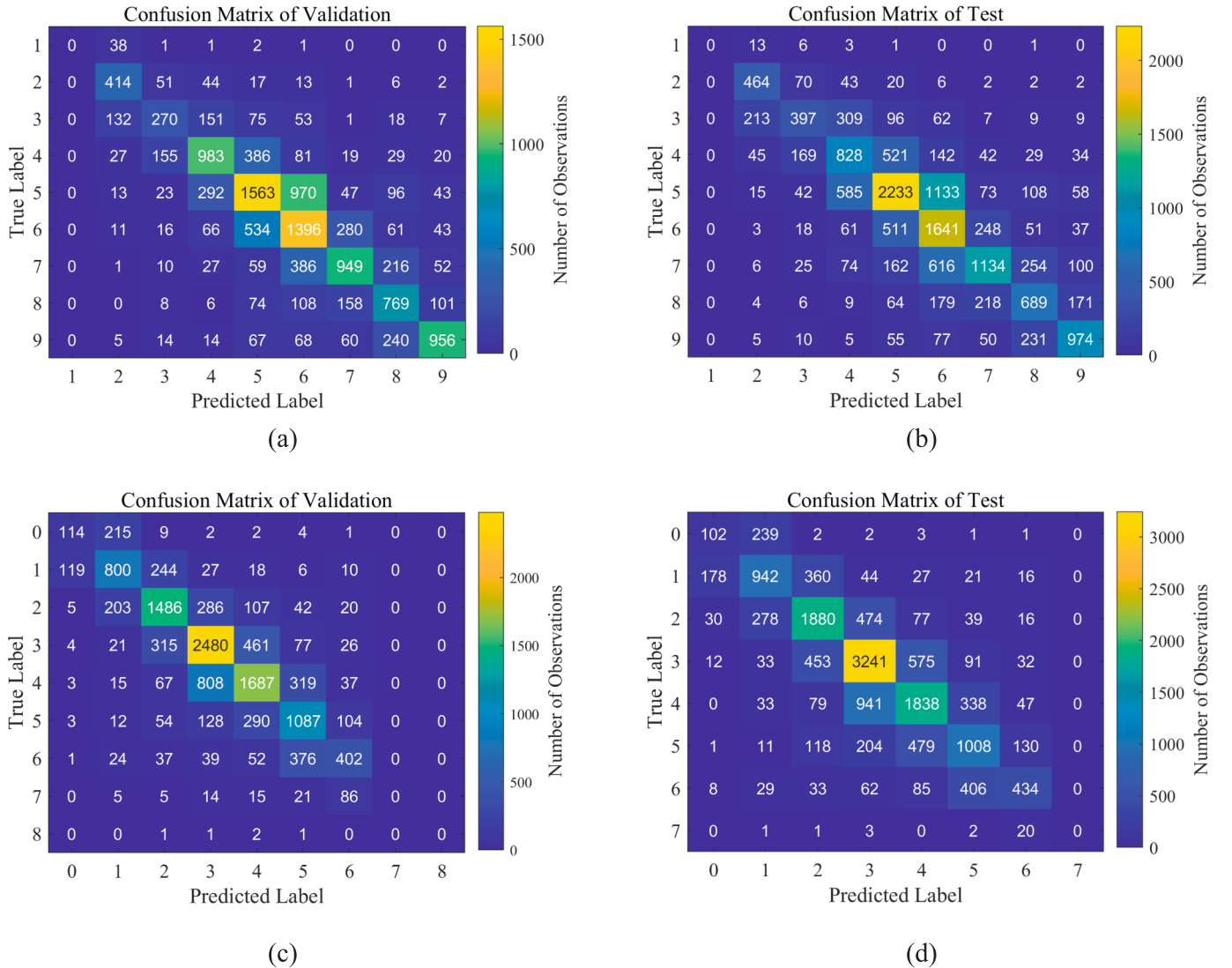


Fig. 5. Confusion Matrices of the Classification Network: (a) SBP Validation Set; (b) SBP Test Set; (c) DBP Validation Set; (d) DBP Test Set.

Table 3
Performance of the proposed model.

	SBP				DBP			
	ME	MAE	RMSE	MAPE	ME	MAE	RMSE	MAPE
Validation	-0.12	4.85	6.77	4.35	0.23	3.66	5.07	4.81
Test	-0.21	4.84	6.74	4.41	0.25	3.46	4.81	4.56

evaluated on both validation and test datasets.

The results demonstrate that the proposed model achieves high accuracy in BP estimation, with low MAE and RMSE. The ME values indicate slight bias in the model's predictions, but these biases are small and manageable. The MAPE further confirms the model's robustness and reliability during both validation and testing phases.

During the 120-epoch training process, both SBP and DBP loss functions (Fig. 6) exhibited stable convergence, with the validation curves plateauing after approximately 100 epochs in SBP training and 90 epochs in DBP training, indicating effective optimization without overfitting.

As shown in Table 4, our model met the accuracy thresholds for BHS Grade A in both SBP and DBP estimation (SBP: 63.9 %/88.1 %/95.3 %; DBP: 77.3 %/94.5 %/98.9 % within 5/10/15 mmHg error bands). Although the cohort size ($n = 30$) is below the BHS-recommended 85 subjects for full validation, these results indicate the method's potential to satisfy clinical-grade accuracy requirements when scaled.

Fig. 7 presents a statistical analysis of radar-based BP estimation, including linear regression fitting plots, Bland-Altman scatter plots, box plots, and error histograms for both SBP and DBP. The linear regression plots (Fig. 7a and e) show a strong correlation between predicted and actual BP values, with regression coefficients $r = 0.93$ and determination coefficients $R^2 = 0.87$ for SBP, $r = 0.94$ and $R^2 = 0.89$ for DBP. These results indicate a high level of predictive accuracy and reliability.

The error distribution analysis reveals that most errors are concentrated within narrow intervals, demonstrating close alignment between estimation and true measurements. Notably, SBP shows a slightly wider error spectrum compared to DBP. The Bland-Altman plots (Fig. 7b and f) further confirm this, with most discrepancies falling within the expected bounds of mean difference ± 1.96 times the standard deviation, underscoring the model's consistency and reliability across the entire measurement range. The box plots (Fig. 7c and g) and error histograms (Fig. 7d and h) provide additional insights, showing that the majority of errors cluster tightly around zero, with only a few outliers. These findings highlight the robustness and accuracy of the proposed model in non-contact BP monitoring.

Table 4
Accuracy performance compared to BHS grade thresholds.

Cumulative Frequency of Error (%)	<5mmHg	<10 mmHg	<15 mmHg
Grade A	60	85	95
Grade B	50	75	90
Grade C	40	65	85
SBP	63.90	88.09	95.30
DBP	77.30	94.46	98.88

5.4. Result of cross-validation protocol

To ensure robust evaluation of our model's generalization capabilities while mitigating subject-specific biases, we implemented a 10-fold cross-validation strategy. This approach maximizes utilization of limited clinical data while maintaining separation between training and testing subjects.

The validation protocol followed these key steps:

- **Subject Partitioning:** The cohort of 30 subjects was randomly divided into 10 mutually exclusive folds using a fixed random seed (seed = 42) to ensure reproducibility. Each fold contained data of 3 subjects, preserving balanced representation across validation sets.
- **Iterative Evaluation:** During each iteration, 9 folds served as training data while the remaining fold constituted the test set.
- **Performance Aggregation:** Model performance metrics (MAE, RMSE) were computed for each test fold and subsequently averaged across all 10 iterations.

As summarized in Fig. 8, the 10-fold cross-validation revealed robust performance metrics across all partitions: For SBP, we observed a MAE of 5.11 ± 0.66 mmHg and RMSE of 6.44 ± 0.78 mmHg, with optimal performance in Fold 2 (MAE = 3.83 mmHg) contrasting with the most challenging Fold 4 (MAE = 6.06 mmHg), while the limited standard deviation ($\sigma < 0.8$ mmHg) across folds confirmed stable generalization across subject cohorts. For DBP, performance metrics showed a positive

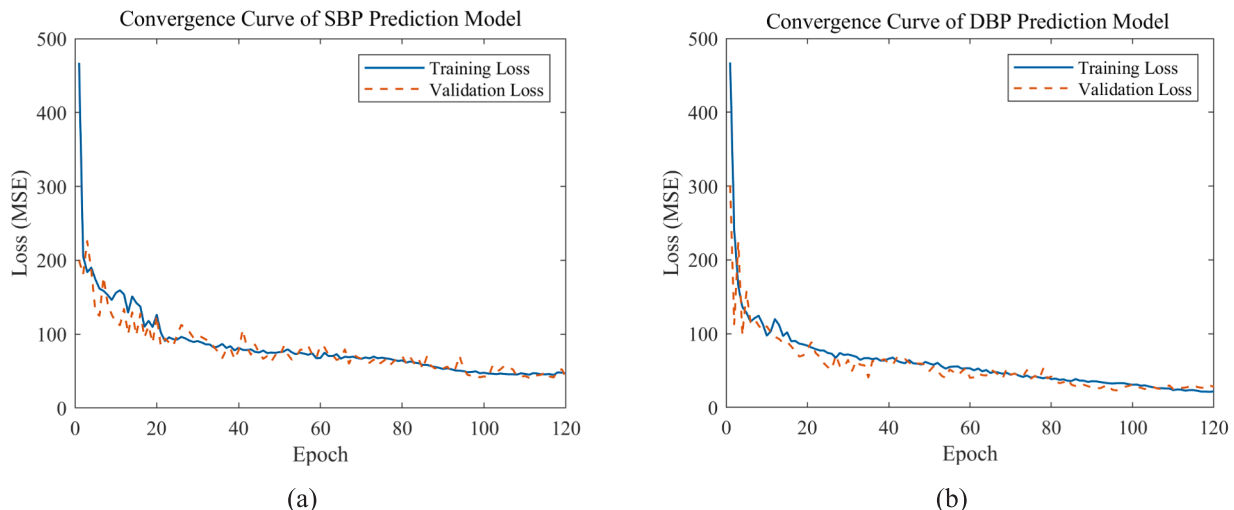


Fig. 6. Convergence Curve of Prediction Model: (a) Convergence Curve of SBP training; (b) Convergence Curve of DBP training.

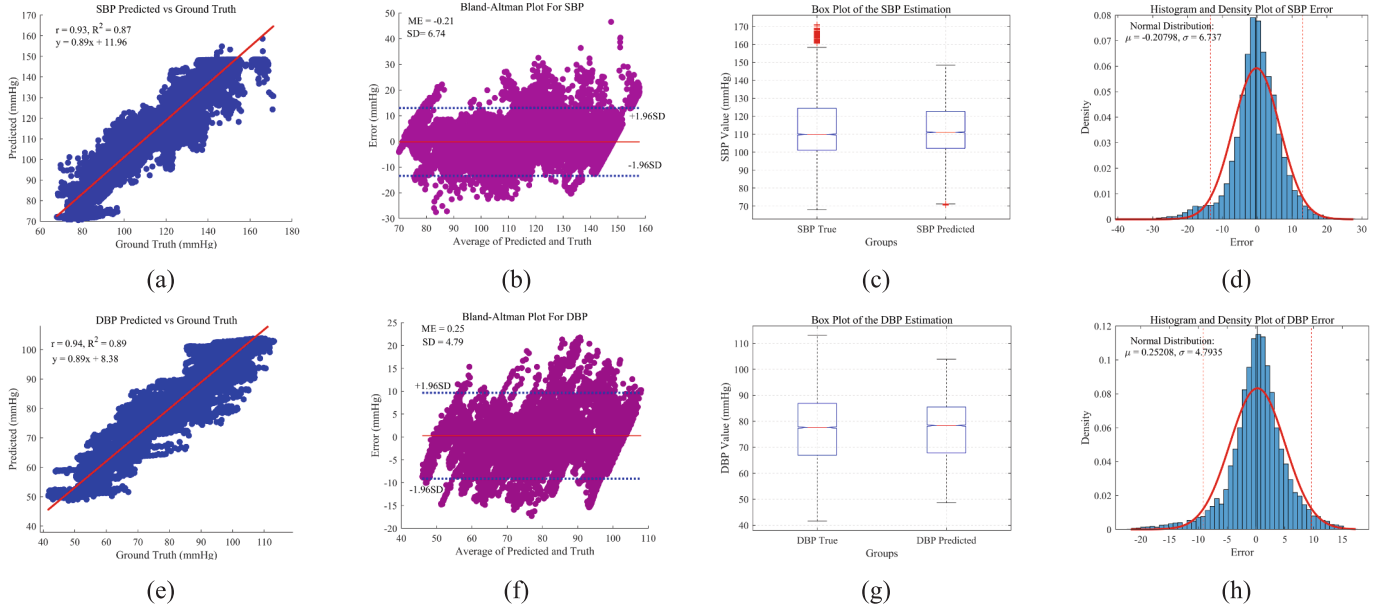


Fig. 7. Statistical Analysis of Radar-Based BP Estimation: (a) Linear Regression Fitting Plot for SBP Estimation; (b) Bland-Altman Scatter Plot for SBP Estimation; (c) Box Plot of SBP Estimation; (d) Error Histogram of SBP Estimation; (e) Linear Regression Fitting Plot for DBP Estimation; (f) Bland-Altman Scatter Plot for DBP Estimation; (g) Box Plot of DBP Estimation; (h) Error Histogram of DBP Estimation.

trajectory in later folds (MAE = 3.91 ± 0.69 mmHg, RMSE = 4.93 ± 0.81 mmHg), culminating in the lowest error at Fold 10 (MAE = 3.37 mmHg), with DBP estimation consistently outperforming SBP by 23.6 % in MAE and 23.4 % in RMSE reduction on average.

Validation Insights: Inter-fold variability remained within clinically acceptable limits, confirming effective mitigation of subject-specific biases through our partitioning strategy, though Folds 4–5 emerged as outliers where both SBP and DBP errors peaked simultaneously, suggesting cohort-specific physiological challenges. The final three folds (8–10) demonstrated progressive DBP improvement, indicating enhanced model adaptation to population diversity, while the MAE-RMSE discrepancy ($\Delta = 1.19$ mmHg for SBP, $\Delta = 1.02$ mmHg for DBP) suggests occasional large errors warranting targeted investigation in outlier cases.

Our subject-wise analysis revealed substantial inter-individual variability in BP estimation accuracy across the 30 participants (Fig. 9). The MAE for SBP ranged from 2.30 mmHg (Subject 24) to 8.57 mmHg (Subject 15), representing a 3.7-fold difference in error magnitude. Similarly, DBP estimation errors varied from 1.95 mmHg (Subject 24) to 5.36 mmHg (Subject 12), demonstrating a 2.7-fold differential. This heterogeneity was particularly pronounced in Subject 15, which exhibited the highest SBP error (MAE = 8.57 mmHg) while simultaneously achieving the third-best DBP performance (MAE = 1.98 mmHg). Such divergent performance patterns suggest subject-specific physiological factors—potentially including vascular compliance variations, motion artifact susceptibility, or sensor-skin interface characteristics—may significantly influence BP estimation accuracy.

The performance of proposed model across 5 physiological states was evaluated in 30 subjects. The results for SBP and DBP are presented in Fig. 10.

For SBP, the resting state exhibited the lowest errors, with a mean MAE of 4.1 ± 1.9 mmHg and mean RMSE of 5.0 ± 2.1 mmHg. In contrast, the apnea state showed the highest errors, with a mean MAE of 6.7 ± 1.8 mmHg and mean RMSE of 8.3 ± 2.4 mmHg. The Valsalva and Tilt up states demonstrated moderate errors. Besides, several outliers were observed, particularly in the apnea state, where the maximum RMSE reached 12.93 mmHg.

For DBP, a similar pattern was observed: the resting state had the lowest errors (mean MAE: 3.0 ± 1.2 mmHg; mean RMSE: 3.7 ± 1.4

mmHg), while the apnea state had the highest errors (mean MAE: 6.2 ± 2.1 mmHg; mean RMSE: 7.1 ± 2.3 mmHg). The performance of the model in the Valsalva and Tilt down states was intermediate. Again, extreme errors in DBP were noted in the apnea state, with the highest RMSE being 10.69 mmHg.

Based on the above results, the inconsistent data availability across physiological states—particularly for apnea, where approximately 20 % of data were missing—may introduce selection bias and reduce statistical power. The limited measurement data in apnea scenarios further restricts data capture during critical pressure transitions, potentially amplifying the error metric.

Secondly, cross-state performance presents a significant challenge to algorithmic generalization. Models exhibited decrease accuracy during autonomic stressors (e.g., Valsalva, apnea), where rapid BP fluctuations exceeded algorithmic tracking capabilities. This limitation highlights the inherent difficulty in maintaining precision across dynamically shifting physiological states.

6. Discussion

6.1. Analysis of classification labeling strategy

The classification labeling strategy critically influences the performance of our dual-task learning architecture, as the discrete category definitions directly impact both the classification task and the subsequent regression refinement. To systematically evaluate this effect, we compared four binning strategies:

- 5 mmHg intervals (24 SBP classes, 19 DBP classes);
- 10 mmHg intervals (13 SBP classes, 10 DBP classes);
- 20 mmHg intervals (7 SBP classes, 5 DBP classes);
- Clinical guidelines-based bins (normal, high-normal, grade 1/2/3 hypertension).

Table 5 compares classification performance across binning strategies and loss functions. With the increase of the box separation width (from 5 mmHg to clinical box separation), the classification accuracy gradually improved (the accuracy of SBP increased from 0.354 for 5 mmHg to 0.775 for clinical box separation). This is because a wider

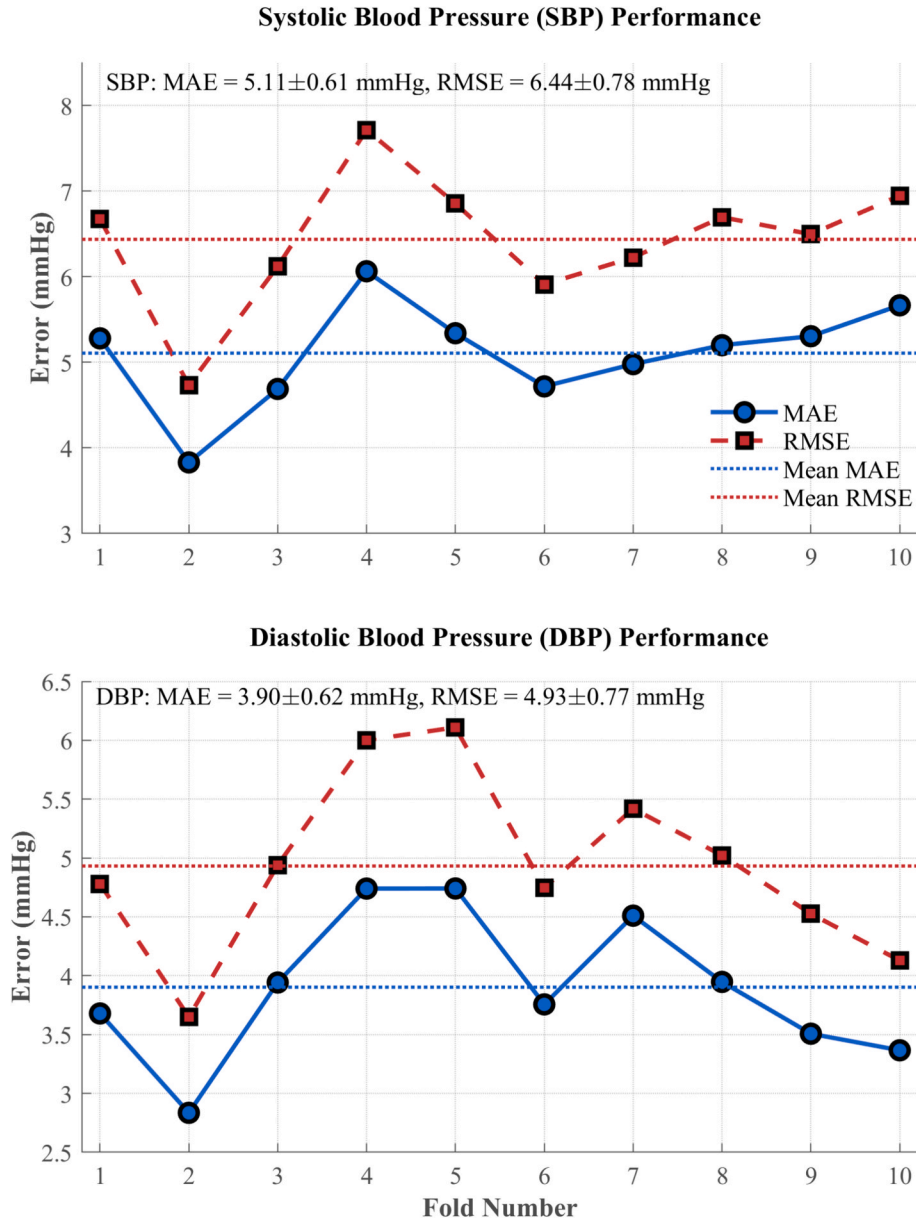


Fig. 8. 10-Fold Cross-Validation Performance: (a) Performance of SBP Estimation; (b) Performance of DBP Estimation.

category covers a larger range of BP, making classification easier. At the same time, the proportion of adjacent categories (the proportion predicted to fall on the adjacent categories of the true category) also decreases as the width of the compartments increases.

Furthermore, we investigated the impact of the category information obtained using different box splitting strategies on the estimation of BP regression, and the results are shown in Table 6.

The regression performance analysis reveals critical insights into the relationship between binning granularity and prediction efficacy. Although wider bin intervals (20 mmHg and clinical bins) improve classification accuracy—as evidenced by the 0.705 SBP accuracy for 20 mmHg bins and 0.775 for clinical bins—they provide diminished utility for regression refinement. The inherent trade-off arises because broader bins sacrifice granular physiological transitions essential for accurate regression.

Notably, 5 mmHg and 10 mmHg bins demonstrate comparable regression accuracy (SBP MAE: 4.83 vs. 4.84 mmHg). However, the 10 mmHg strategy proves superior by balancing three key factors:

- **Model Efficiency:** With 13 SBP and 10 DBP classes, it reduces category counts by 42 % compared to 5 mmHg bins (24 SBP, 19 DBP classes), streamlining architecture without sacrificing precision.
- **Clinical Alignment:** Studies have shown that every 10 mmHg decrease in SBP is significantly associated with a reduction in major cardiovascular events and all-cause mortality [42], and the main fluctuation range of BP measurement affected by external factors such as the environment is within the range of 10–20 mmHg [43].
- **Technical standards:** The AAMI/ESH/ISO validation protocol requires ≥ 85 % of predictions to fall within 10 mmHg error tolerance [44], making the 10 mmHg bin width inherently aligned with measurement variability.

This tripartite synergy enables the 10 mmHg strategy to preserve continuous hemodynamic dynamics while mitigating measurement noise, and achieve the equilibrium: regression stability, clinical actionability, and computational efficiency.

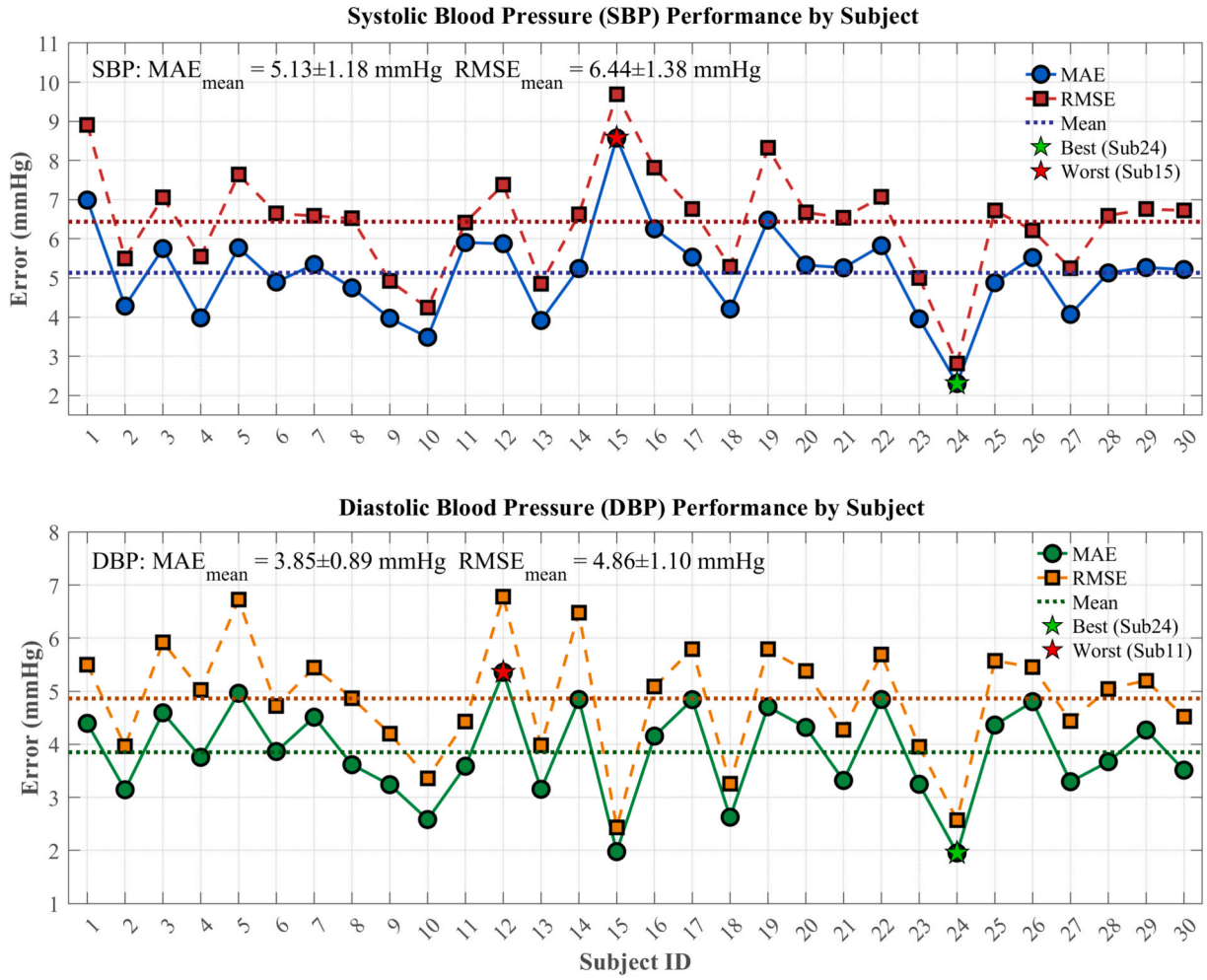


Fig. 9. Individual Subject Performance Analysis:(a) Performance of SBP Estimation; (b) Performance of DBP Estimation.

6.2. Effectiveness of the custom loss function

The custom loss function (Eqs. (7)–(10)) may address a potential challenge in BP classification: the continuous nature of BP could create inherent ambiguity near bin boundaries (e.g., 129 mmHg vs. 131 mmHg). Traditional cross-entropy loss might equally penalize all non-exact class predictions, potentially overlooking physiological contexts where adjacent categories (e.g., 120–130 mmHg and 130–140 mmHg) could contain clinically comparable information. Our solution include an adaptive reward mechanism: This may provide partial credit ($\alpha = 0.5$) for predictions within ± 10 mmHg of true values, potentially reflecting clinical tolerance where 142 mmHg could be considered diagnostically similar whether classified as 130–140 or 140–150 mmHg. This design acknowledges the possibility that radar signals near BP transitions might exhibit hybrid characteristics—for example, a 132 mmHg signal could share more features with 128 mmHg (adjacent bin) than 138 mmHg (same bin), suggesting flexible error tolerance might be beneficial.

As indicated in Table 6, the custom loss appears to yield performance improvements, particularly where physiological ambiguity is likely highest:

Transition zone recognition: For 10 mmHg bins—a clinically relevant granularity—recognition ratio increased by $\sim 3.17\%$ ($0.853 \rightarrow 0.88$ SBP), potentially correlating with regression improvements (RMSE reduced by $\sim 1.02\%$ in Table 6).

Multi-task interactions: Classification outputs might serve as informative priors for regression, possibly contributing to variance reduction.

This could help explain the observed MAE improvement despite identical binning strategies.

These benefits tended to be most pronounced with 10 mmHg bins—a strategy aligning with natural BP variability (± 5 –8 mmHg). Finer bins (5 mmHg) showed more modest gains (small adjacent ratio increase), potentially due to noise sensitivity, while coarser bins (20 mmHg) exhibited minimal improvements (marginal adjacent ratio decrease), possibly indicating insufficient transition resolution. This suggests the loss function might operate optimally when bin width approximates clinically meaningful BP change thresholds.

Collectively, the custom loss could help reframe classification from a categorical task toward a feature refinement process. By accommodating physiologically plausible errors, it may capture transitional patterns that regression alone might overlook and enhancing robustness to signal artifacts. This approach demonstrates one possible way to incorporate domain knowledge into loss design, potentially bridging algorithmic and clinical considerations.

6.3. Effectiveness of classification-information fusion in regression prediction: Comparison with baseline methods

To evaluate the performance of the proposed approach, we compared it with several baseline models, including typical waveform feature extraction methods (CNN and ResNet), temporal dependency extraction networks (LSTM and GRU), and methods that combine spatial and temporal features, such as ResNet + LSTM, ResNet + BiLSTM, ResNet + GRU, and ResNet + BiGRU. The results are summarized in

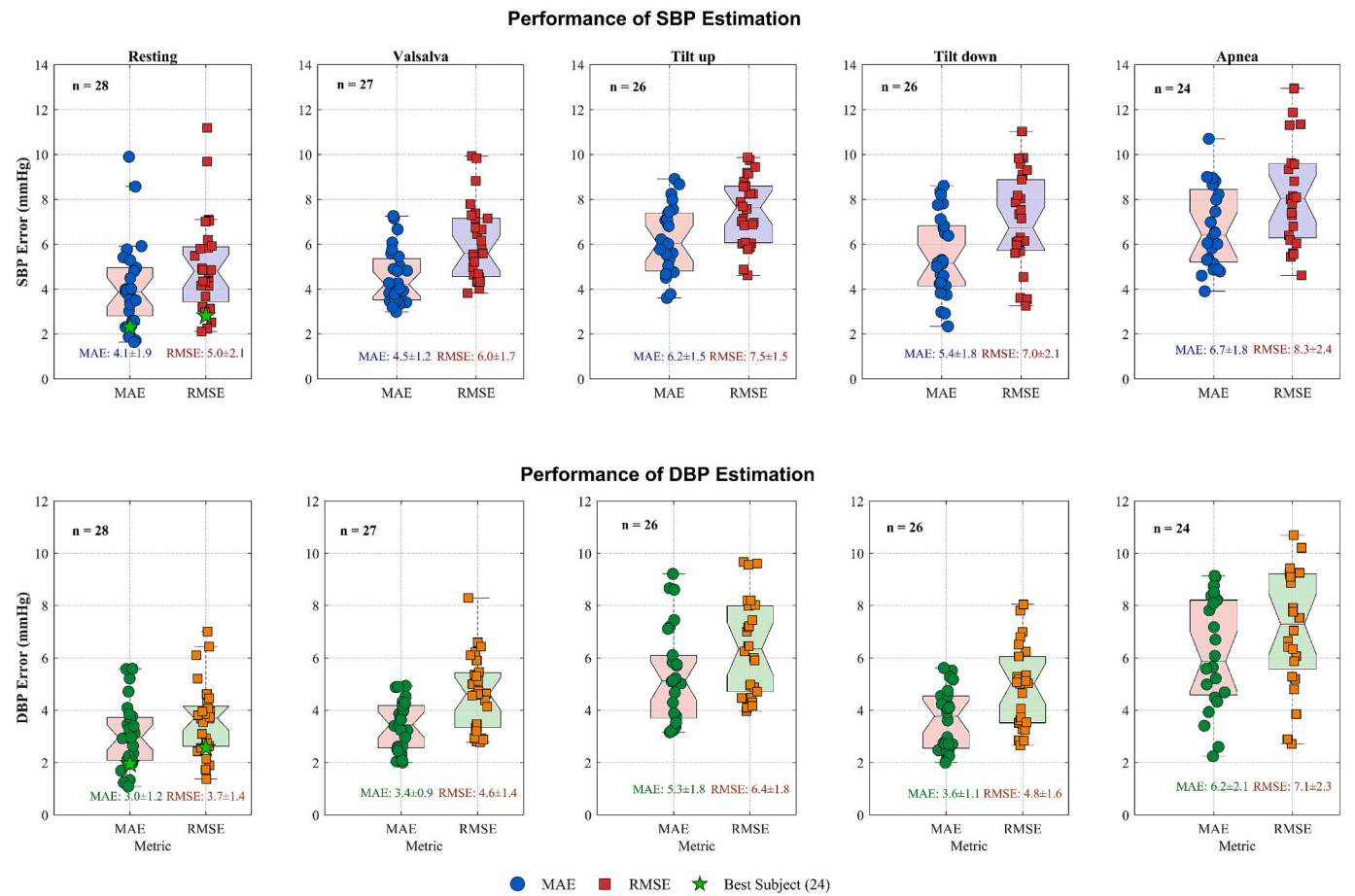


Fig. 10. Physiological State-Dependent Performance Analysis.

Table 5
Compares classification performance across binning strategies and loss functions.

Binning Strategy	Loss Function	SBP			DBP		
		Accuracy	Adjacent Ratio	Total ratio	Accuracy	Adjacent Ratio	Total ratio
5 mmHg	Cross-entropy	0.354	0.341	0.695	0.4316	0.3766	0.808
	Custom	0.353	0.36	0.713	0.426	0.389	0.815
10 mmHg	Cross-entropy	0.532	0.321	0.853	0.606	0.308	0.914
	Custom	0.541	0.339	0.88	0.610	0.315	0.925
20 mmHg	Cross-entropy	0.705	0.248	0.953	0.720	0.255	0.975
	Custom	0.720	0.239	0.959	0.721	0.257	0.978
Clinical	Cross-entropy	0.7746	0.1963	—	0.727	0.209	0.936

*Total ratio = Accuracy + Adjacent Ratio.

Table 6
Regression performance under different binning strategies and loss function.

Binning Strategy	Loss Function	SBP		DBP	
		MAE	RMSE	MAE	RMSE
5 mmHg	Cross-entropy	4.93	6.85	3.50	4.94
	Custom	4.83	6.75	3.43	4.80
10 mmHg	Cross-entropy	4.88	6.81	3.44	4.86
	Custom	4.84	6.74	3.46	4.81
20 mmHg	Cross-entropy	4.85	6.91	3.48	4.88
	Custom	4.87	6.93	3.49	4.91
Clinical	Cross-entropy	4.85	6.86	3.47	4.88

Table 7.

The results clearly demonstrate that single-feature extraction networks, such as CNN for spatial information and LSTM and GRU for temporal information, perform poorly in radar signal-based BP prediction. Although the multi-scale ResNet improves BP estimation accuracy, it still falls short of meeting practical requirements.

Significant improvements are observed when spatial and temporal features are combined. Models such as ResNet + LSTM, ResNet + BiLSTM, ResNet + GRU, and ResNet + BiGRU show substantial enhancements across all evaluation metrics. Building on these findings, our proposed model utilizes a cross-entropy loss function with a multi-scale ResNet to extract classification probability features. These features are then integrated with temporal and waveform features, providing additional perspectives and information for BP prediction.

Table 7

Comparison of BP estimation performance using different methods.

Method	Dataset	SBP (mmHg)				DBP (mmHg)			
		ME	MAE	RMSE	MAPE	ME	MAE	RMSE	MAPE
CNN	Validation	1.59	9.59	13.30	8.62	0.05	7.29	9.75	9.97
	Test	1.02	9.80	13.12	8.79	0.14	7.25	9.79	9.88
ResNet	Validation	-0.95	6.09	9.55	5.39	-0.51	4.42	7.06	5.69
	Test	-0.94	6.21	9.37	5.56	-0.38	4.59	6.96	6.00
GRU	Validation	0.15	14.02	17.64	12.84	0.13	10.69	13.52	14.67
	Test	0.69	14.10	17.62	12.97	0.63	10.61	13.40	14.73
LSTM	Validation	1.41	13.99	17.56	12.94	1.40	10.67	13.38	14.84
	Test	0.39	13.95	17.73	12.73	0.84	10.57	13.29	14.58
ResNet + LSTM	Validation	0.68	5.38	7.49	4.89	-0.07	3.64	5.47	4.74
	Test	-0.45	5.37	7.51	4.81	-0.04	3.83	5.51	5.05
ResNet + BiLSTM	Validation	-0.24	5.01	7.06	4.41	-0.48	3.61	5.44	4.63
	Test	-1.32	5.01	7.04	4.44	-0.36	3.78	5.38	4.92
ResNet + GRU	Validation	0.18	5.18	7.32	4.70	0.07	3.57	5.38	4.64
	Test	0.42	5.14	7.24	4.66	-0.26	3.79	5.41	4.99
ResNet + BiGRU	Validation	-0.06	5.07	7.22	4.57	0.14	3.54	5.15	4.58
	Test	0.19	5.10	7.17	4.60	0.25	3.75	5.25	4.93
Dual-task learning model	Validation	-0.12	4.85	6.77	4.35	0.23	3.66	5.07	4.81
	Test	-0.21	4.84	6.74	4.41	0.25	3.46	4.81	4.56

The integration of classification-derived features into the regression framework demonstrates measurable improvements in BP estimation accuracy when compared to the baseline ResNet + BiGRU architecture. As evidenced in Table 7, the dual-task learning architecture consistently outperformed the non-fusion approach across critical error metrics in both validation and test datasets:

For SBP estimation: MAE decreased by 4.3 % (5.07 vs 4.85 mmHg) in validation and 5.1 % (5.10 vs 4.84 mmHg) in test data; RMSE reduced by 6.2 % (7.22 vs 6.77 mmHg) in validation and 6.0 % (7.17 vs 6.74 mmHg) in test data.

For DBP estimation: Test MAE improved by 7.7 % (3.75 vs 3.46 mmHg); Test RMSE decreased by 8.4 % (5.25 vs 4.81 mmHg).

Notably, the dual-task learning architecture achieved these enhancements while maintaining comparable mean error (ME) values, suggesting the classification-derived features primarily reduce estimation variance rather than introducing systematic bias.

These improvements may stem from the classification branch's ability to provide contextual constraints to the regression network. By first categorizing BP ranges (e.g., identifying 120–130 mmHg intervals), the model potentially establishes physiological boundaries that guide finer-grained regression predictions, reducing outlier errors that disproportionately impact RMSE. The stable performance across validation and test sets further suggests the fusion architecture mitigates overfitting risks inherent in pure regression approaches.

6.4. Comparative results with related work

Existing studies on non-contact continuous BP estimation using radar technology vary considerably in terms of radar modes, sample sizes, and

experimental settings, making direct comparisons difficult. Studies with a limited number of subjects or samples often exhibit narrower BP prediction ranges, which can lead to inflated performance metrics. As such, our comparative analysis provides a fundamental performance benchmark for various non-contact BP estimation techniques.

As summarized in Table 8, our method achieves competitive performance under continuous BP labels and diverse physiological states (resting, Valsalva, breath-holding, tilt-table tests). Key observations include:

- Our DBP MAE (3.46 mmHg) and RMSE (4.81 mmHg) outperform all comparable methods under continuous labeling and multi-state conditions, including SDCN [31] (DBP MAE = 4.42, RMSE = 5.5) and U-net + RF [32] (DBP MAE = 3.82, RMSE = 5.81). This suggests enhanced robustness to hemodynamic fluctuations during dynamic maneuvers. While SBP MAE (4.84 mmHg) matches U-net + RF [32], it slightly outperform SDCN [31] (4.84 vs. 4.86).
- Static vs. Dynamic Performance: Methods like ResNet [30] (MAE_{SBP} = 3.51, MAE_{DBP} = 2.44) excel in resting states but lack validation under motion – highlighting a critical trade-off between precision and generalizability.
- Impact of BP Labeling and Real-Time Constraints: Discrete labels (e.g., [22,29,30]) derived from cuff-based measurements inherently limit temporal resolution (≥ 2 -min intervals), restricting their clinical utility for continuous monitoring. Despite similar continuous labeling, [27] and [28] report higher errors (e.g., [27]: DBP RMSE = 11.0) due to simplified scenarios (resting/2-state) and insufficient temporal modeling. Our approach uses continuous labels with a 2-second

Table 8

Statistical comparative evaluation of model performance.

Year	Method	Subject	BP Labels	Scenarios	SBP (mmHg)				DBP (mmHg)			
					ME	MAE	RMSE	MAPE	ME	MAE	RMSE	MAPE
2022	NN [22]	27	Discrete	3 states	–	9.51	–	–	–	6.69	–	–
2023	NN [27]	55	Continuous	Resting	9.2	–	11.0	–	7.7	–	11.0	–
2023	LR [28]	27	Continuous	2 states	–	5.54	7.62	–	–	4.68	6.15	–
2023	CNN + GRU [29]	36	Discrete	Resting	–	4.74	5.52	–	–	4.49	5.25	–
2024	ResNet [30]	100	Discrete	Resting	0.13	3.51	6.90	–	-0.02	2.44	4.61	–
2024	SDCN [31]	30	Continuous	5 states	-0.32	4.86	6.14	4.40	-0.2	4.42	5.5	5.69
2025	U-net + RF [32]	25	Continuous	5 states	–	4.84	7.51	–	–	3.82	5.81	–
2024	Ours	30	Continuous	5 states	-0.21	4.84	6.74	4.41	0.25	3.46	4.81	4.56

MLP: Multilayer Perceptron; RF: Random Forest; NN: Neural Network; LR: Linear Regression; SDCN: Stacked Deformable Convolution Network; ResNet: Residual network.

estimation interval, enabling near-real-time BP tracking crucial for detecting rapid changes during tilt-table tests or Valsalva maneuvers.

This advantage stems from two architectural innovations: First, the spatial feature extractor combines large-kernel initial convolutions (kernel = 15) with hierarchical residual blocks (4-layer BasicBlock), enhancing pulse waveform characterization while maintaining simpler topology than SDCN's deformable convolutions in [31]. Second, the temporal module employs a triple-layer BiGRU with progressive dimensionality reduction, enabling deeper state-transition modeling than single-layer GRU designs in [29]. The architecture's complexity is functionally justified: The residual blocks capture nuanced morphological shifts during hemodynamic stress, while the deep BiGRU stack decodes prolonged physiological responses. Simultaneously, the classification branch utilizes cross-entropy loss to generate BP range probabilities, providing supplementary regularization for the regression backbone to constrain outlier contributions.

Notably, our design prioritizes physiological alignment over parameter minimization. While direct architectural comparisons remain infeasible due to incomplete disclosures in existing literature, the framework's efficacy and the strategy of leveraging classification-derived probability insights are empirically validated. This classification-derived probabilistic regularization approach holds promise for extending to other BP estimation architectures to enhance performance ceilings.

6.5. Limitations and future work

Despite achieving high-precision, contactless BP monitoring using chest radar echo signals and the proposed model, several challenges remain:

Inter-subject Variability in Estimation Accuracy: Substantial performance differences persist across individuals (e.g., DBP MAE range: 1.97–5.36 mmHg). This heterogeneity may reflect unmodeled physiological factors such as vascular compliance variations or tissue attenuation characteristics. While our feature extractor captures general hemodynamic patterns, its capacity for subject-specific adaptation remains limited without incorporating biometric covariates (e.g., arterial stiffness indices). To address inter-subject variability, several promising pathways may be explored: Integrating biometric metadata (e.g., baseline BP, vascular stiffness indices) as auxiliary inputs could enhance personalized feature adaptation. Expanding datasets through multi-center collaborations would further improve robustness across physiological diversities, potentially mitigating current accuracy fluctuations between individuals.

Divergent SBP-DBP Estimation Performance: The systematic dissociation between systolic and diastolic accuracy indicates incomplete modeling of cardiovascular coupling mechanisms. Subject 15's contradictory results—high SBP error with exceptional DBP precision—demonstrate that the current architecture processes pressure phases independently rather than exploiting their pathophysiological interdependence. The paradoxical performance observed in Subject 15 suggests value in developing coupled estimation mechanisms—potentially through joint hemodynamic constraints that enforce physiological relationships between systolic and diastolic phases. Regarding SBP-DBP estimation discrepancies, future efforts could focus on modeling cardiovascular interdependence more explicitly.

Cross-State Generalization Challenge: Performance degradation during dynamic maneuvers (Resting vs. Tilt up test: Δ SBP MAE = +2.1 mmHg) reveals two probable constraints: First, motion artifacts introduce non-stationary noise that exceeds the temporal module's denoising capacity. Second, the absence of explicit state-detection mechanisms forces the BiGRU to model physiological transitions implicitly—a task complicated by inter-state feature distribution shifts. This suggests that current architectural expressiveness may be sufficient for stable conditions but inadequate for abrupt hemodynamic changes. For cross-state

generalization challenges, two complementary approaches appear worth investigating: Implementing trainable state-transition detectors could allow dynamic reconfiguration of temporal weights during physiological shifts, while adversarial augmentation synthesizing radar-specific motion artifacts might enhance noise resilience.

Complexity of Dual-Task Regularization: While classification-derived probability constraints improve regression robustness, the enforced feature separation between tasks increases architectural overhead. Maintaining distinct feature extractors for classification (BP ranges) and regression (continuous values) introduces approximately 30 % additional parameters to single-task equivalents. Subsequent research could explore architectural refinements to better balance the tradeoffs between performance gains and computational overhead. Besides, clinical adoption requires validation per BHS-recommended subjects (≥ 85 subjects), and we still need to further expand the experimental dataset to validate the algorithm.

7. Conclusion

This study advances non-contact continuous BP monitoring through a dual-task learning model that integrates spatial feature extraction, temporal dependency modeling, and physiological regularization. The framework combines a multi-scale ResNet backbone with hybrid convolutional kernels (15/7/3) and a triple-layer BiGRU architecture, capturing cardiovascular continuity patterns through sequential pulse correlation. This approach enables 2-second-interval monitoring with clinically relevant accuracy: DBP estimation reached ME = 0.25 ± 4.81 mmHg and MAE = 3.46 mmHg, while SBP achieved ME = -0.21 ± 6.74 mmHg and MAE = 4.84 mmHg—accuracy satisfying British Hypertension Society Grade A standards **despite cohort size limitations**.

The classification-regularized regression framework contributes physiological context through BP range priors (10 mmHg bins), helping reduce estimation variance by 8.4 % in DBP RMSE (4.81 vs. 5.25 mmHg) compared to regression-only baselines. This integration appears to suppress physiologically implausible outputs during state transitions while maintaining 0.5 Hz sampling resolution. Cross-validation across resting, apnea, and autonomic stressors scenario demonstrated consistent performance.

These innovations collectively contribute to addressing core challenges in non-contact hemodynamic sensing, potentially mitigating motion artifact sensitivity and physiological nonstationarity effects. Future efforts will explore radar-sensor fusion for enhanced motion robustness, architectural refinements for efficient deployment, and expanded validation with diverse populations. The methodology represents a meaningful step toward practical contactless monitoring solutions, where physiological-informed learning may help bridge technical innovation and clinical application.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used tongyi.ai and deepseek.com in order to improve language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRedit authorship contribution statement

Pengfei Wang: Writing – original draft, Software, Funding acquisition. **Peilin Shu:** Writing – review & editing, Formal analysis. **MingHao Yang:** Methodology. **Hongqiu Zhang:** Methodology, Formal analysis. **Jianqi Wang:** Writing – review & editing, Conceptualization. **Cong Wang:** Writing – review & editing, Funding acquisition. **Hongbo Jia:** Project administration, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A:. Model architectures and hyperparameter configurations

A.1 Model architecture specifications

The proposed dual-task framework comprises two core components: a classification network and a regression network. The classification network employs a modified ResNet architecture for BP range classification, while the regression network adopts a hybrid ResNet-BiGRU design for continuous blood pressure prediction. Both networks share identical feature extraction backbones (ResNet modules) but incorporate task-specific output heads.

A.1.1 Classification network architecture

The classification network (Table A1) features an end-to-end ResNet architecture optimized for BP range recognition. Key innovations include:

- Dual-scale convolutional design: 15- and 7-kernel convolutions capture long-range trends and local features;
- Residual connection optimization: Four-stage residual structure (64–128–256–512 channels) with skip connections mitigates gradient vanishing;
- Adaptive output: Global average pooling enables variable-length input handling;
- Regularization mechanism: Dropout layer before classification head prevents overfitting.

Table A1
Classification network architecture.

Layer/Block	Operation and Configuration	Output Size
Input Layer	Single-channel input signal	(batch, 1, 300)
<i>Initial Convolution</i>		
Conv1d	kernel_size = 15, stride = 2, padding = 3, filters = 64, ReLU, BatchNorm	(batch, 64, 146)
MaxPool1d	kernel_size = 3, stride = 2, padding = 1	(batch, 64, 73)
Conv1d	kernel_size = 7, stride = 2, padding = 3, filters = 64, ReLU, BatchNorm	(batch, 64, 36)
MaxPool1d	kernel_size = 3, stride = 2, padding = 1	(batch, 64, 18)
<i>Residual Blocks</i>		
Layer1 (2 × BasicBlock)	64 filters, kernel_size = 3, stride = 1	(batch, 64, 18)
Layer2 (2 × BasicBlock)	128 filters, stride = 2	(batch, 128, 9)
Layer3 (2 × BasicBlock)	256 filters, stride = 2	(batch, 256, 5)
Layer4 (1 × BasicBlock)	512 filters, stride = 2	(batch, 512, 3)
GlobalAvgPool	Adaptive average pooling	(batch, 512, 1)
flatten	–	(batch, 512)
dropout	Dropout, p = 0.35	(batch, 512)
FC	in = 512, out = num_classes	(batch, num_classes)

A.1.2 Regression Network Architecture

The regression network (Table A2) integrates ResNet feature extraction with BiGRU temporal modeling. Critical innovations include:

- Feature-temporal decoupling: ResNet extracts spatial features, BiGRU models temporal dependencies;
- Bidirectional processing: Three-layer bidirectional GRU captures forward/backward physiological dependencies;
- Multimodal feature fusion: Concatenation of classification features enhances predictive capability;
- Hierarchical regularization: Batch normalization and dropout at multiple stages.

Table A2
Regression network architecture.

Layer/Block	Operation and Configuration	Output Size
Input Layer	Single-channel input signal	(batch, 1, 300)
<i>Initial Convolution</i>		
Conv1d	kernel_size = 15, stride = 2, padding = 3, filters = 64, ReLU, BatchNorm	(batch, 64, 146)
MaxPool1d	kernel_size = 3, stride = 2, padding = 1	(batch, 64, 73)
Conv1d	kernel_size = 7, stride = 2, padding = 3, filters = 64, ReLU, BatchNorm	(batch, 64, 36)

(continued on next page)

Table A2 (continued)

Layer/Block	Operation and Configuration	Output Size
MaxPool1d	kernel_size = 3, stride = 2, padding = 1	(batch, 64, 18)
<i>Residual Blocks</i>		
Layer1 (2 × BasicBlock)	64 filters, kernel_size = 3, stride = 1	(batch, 64, 18)
Layer2 (2 × BasicBlock)	128 filters, stride = 2	(batch, 128, 9)
Layer3 (2 × BasicBlock)	256 filters, stride = 2	(batch, 256, 5)
Layer4 (1 × BasicBlock)	512 filters, stride = 2	(batch, 512, 3)
GlobalAvgPool	Adaptive average pooling	(batch, 512, 1)
<i>BiGRU Layers</i>		
Feature reshape	Flatten, Feature reshape, timesteps = 40	(batch, 40, 512)
BiGRU Layer1	hidden_size = 64 (bidirectional), input_size = 512, BatchNorm1d	(batch, 40, 128)
BiGRU Layer2	hidden_size = 48 (bidirectional), BatchNorm1d	(batch, 40, 96)
BiGRU Layer3	hidden_size = 32 (bidirectional), BatchNorm1d	(batch, 40, 64)
Dropout	rate = 0.35	(batch, 40, 64)
<i>Output Layers</i>		
FC1	Linear(64 + num_classes → 32), ReLU	(batch, 40, 32)
Output	Linear(32 → 1)	(batch, 40, 1)

A.2 Hyperparameter configurations

Tables A3 and A4 detail hyperparameter configurations for classification and regression tasks, respectively. The hyperparameter configurations reflect a unified design philosophy that prioritizes architectural consistency while accommodating task-specific requirements. Both tasks share identical core specifications for the feature extraction backbone, optimization framework, and regularization strategies. Strategic adaptations are implemented where task demands differ: the regression network incorporates temporal modeling capabilities through its BiGRU layers and feature fusion mechanism, while the classification network optimizes its output layer for discrete state discrimination. Both configurations integrate physiological constraints within their loss functions, embedding domain knowledge directly into the learning process.

Table A3
Classification Task Hyperparameters.

Hyperparameter Category	Parameter	Value	Description
Architecture	Input channels	1	Single-channel radar signal
	Input length	300	Features per sample
	Residual block config	[1,2,2,2]	Blocks per layer
	Num classes	Calculated by 10 mmHg binning	BP ranges
Optimization	Learning rate	0.0009	Adam optimizer
	Weight decay	0.001	L2 regularization
	Batch size	64	Training batches
	Algorithm	Adam	
Scheduling	LR Scheduler	StepLR	Learning rate scheduler type
	Step Size	10	Steps between LR adjustments
	Gamma	0.9	LR multiplication factor per step
Training	Epochs	120	Maximum training cycles
	Early stopping patience	20	Validation loss patience
Regularization	Dropout rate	0.3	Before classifier
Loss Function	Base Loss	CrossEntropy	Primary loss function
	Adjacent Reward	0.6	Reward for adjacent class predictions

Table A4
Regression task hyperparameters.

Hyperparameter category	Parameter	Value	Description
Architecture	Input channels	1	Single-channel radar signal
	Input length	300	Features per sample
	Residual block config	[1,2,2,2]	Blocks per layer
	BiGRU hidden layers	[64,48,32]	Units per layer (unidirectional)
	Fusion dimension	64 + N_bins	BiGRU output + BP bin classes

(continued on next page)

Table A4 (continued)

Hyperparameter category	Parameter	Value	Description
Optimization	Learning rate	0.0009	Adam optimizer
	Weight decay	0.001	L2 regularization
	Batch size	32	Training batches
	Optimization algorithm	Adam	
Scheduling	LR Scheduler	StepLR	Learning rate scheduler type
	Step Size	10	Steps between LR adjustments
	Gamma	0.9	LR multiplication factor per step
Training	Epochs	120	Maximum training cycles
	Early stopping patience	20	Validation loss patience
Regularization	Dropout rate	0.35	Before classifier
Loss Function	Base Loss	MSE	Primary loss function
	Gradient clipping	1.0	Max gradient norm

Data availability

Data will be made available on request.

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